

Mithilesh Vaidya

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Education

- **Georgia Institute of Technology**, Atlanta, USA [2022 - 2024]
 - Master's in Computer Science (Specialisation: Machine Learning)
- **Indian Institute of Technology Bombay**, Mumbai, India [2017 - 2022]
 - B.Tech+M.Tech in Electrical Engineering (**CPI 9.52/10**) with a Minor in Computer Science and Engineering

Professional Experience

- **AI Researcher** *June'24 - Present*
Descript, San Francisco *Research Engineer*
 - Co-led **Audio Regenerate**, a latent-inpainting model for seamless word-level audio editing, by training a continuous neural codec and a flow-matching generator
 - Developed **Video Regenerate**, an audio-driven lip-sync model for video editing and translation, and a contrastive model for audio-video alignment
 - Co-authored **PoDAR**, a power-disentangled audio representation that accelerates generator convergence by **2×**
- **LLM-powered Natural Language Interface** *May'23 - Aug'23*
Qualcomm, San Diego | Guide: Mr. Vasudev Nayak, Principal Engineer *Machine Learning Internship*
 - Designed an end-to-end hierarchical modular pipeline for structured parsing using LLMs such as MPT-7B
 - Developed techniques for reducing bias and hallucinations and compared performance across various open-source LLMs
 - Evaluated performance of NVIDIA Nemo and OpenAI Whisper for STT and machine translation
- **Verification of FPGA-based High Frequency Trading Platform** *Apr'20 - June'20*
APT Portfolio Pvt. Ltd. | Guide: Mr. Vivek Pannikar, Senior Verification Engineer *Hardware Engineering Internship*
 - Implemented **Direct Programming Interface**, a protocol for exchanging data between SystemVerilog and C, for **speeding up verification** of testbenches using Cocotb, Quartus and Riviera by **3×**
 - Used Python **metaclasses** for automatically generating Python, SystemVerilog and C DPI header and implementation files from high-level JSON inputs
- **Autonise AI** *Sep'18 - May'19*
Machine Learning Startup *Co-Founder*
 - Delivered an **end-to-end solution** consisting of an annotation tool for labelling datasets, a UNet-based model for **segmenting** out spots, patches and wrinkles in facial images and an AWS API for demonstrating the model
 - Implemented **PixelLink** and a GRU for word-level text detection, **invariant** to font size, colour, background, orientation, etc. and presented it to HDFC bank for automatic field extraction from documents like Aadhar Card, Passport, Driving Licence, etc.

Projects and Publications

- **Prominence Detection and Oral Reading Fluency of Children's Read Speech** *June'21 - June'22*
Guide: Prof. Preeti Rao, IIT Bombay | [arXiv:2110.14273](https://arxiv.org/abs/2110.14273) *Master's Thesis*
 - Replaced a Random Forest Classifier baseline with a **CRNN** framework and Sinc convolution for predicting the degree of prominence for each word in children's read speech
 - Exploited phrase boundary labels in various **multi-task learning** paradigms and POS tags and BERT embeddings for incorporating complementary lexical information
 - Implemented a **Wav2vec2.0**-based end-to-end model for oral reading fluency assessment and outperformed RFC baseline operating on hand-crafted features by **0.06** (absolute Pearson)
- **Switching Subspace Model for Neural Population Analysis** *Sep'22 - May'24*
Guide: Prof. Anqi Wu, GeorgiaTech *Research project*
 - Developed a sequential VAE to capture multiple latent dynamics and introduced a unimodal Gaussian Process prior for interpretability

Honors and Awards

- **Institute Silver Medal** at IIT Bombay for best academic standing in department (**2/70**) [2022]
- **Undergraduate Research Award (URA03)** at IIT Bombay for outstanding thesis contributions [2022]
- **Institute Special Mention** for Journalism carried out as part of Insight, IIT Bombay's student media body [2022]

Key Coursework

- **Machine Learning:** Automatic Speech Recognition, Advanced ML, Foundations of Intelligent and Learning Agents
- **Computer Science:** Data Structures and Algorithms, Computer Graphics, Network Security, Operating Systems